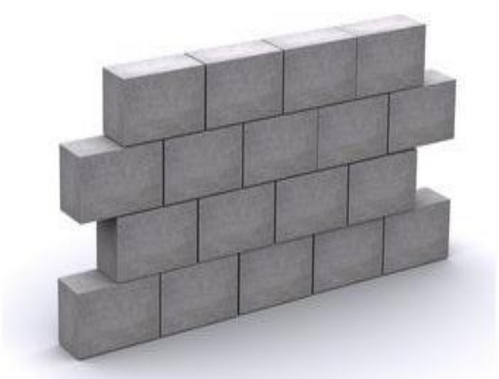
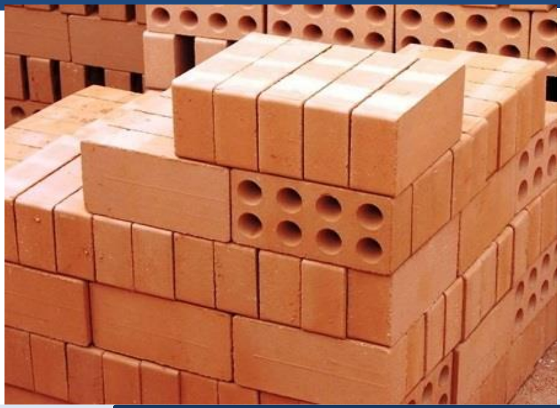
JORDAN
Investment Commission
الأردن
هيئة الاستثمار

A Pre-Feasibility for Establishing an Automatic Bricks Factory In Al-Karak


2017

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JORDAN INVESTMENT COMMISSION هيئة الاستثمار الأردن

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1. Executive Summary

The idea of the project is to establish an automatic brick factory that produces bricks automatically through machines mixing cement, sand and other industrial materials with water in measured proportions, pressed and molded to produce bricks used in construction.

The project will be established in Karak governorate, which is located in the south of the Kingdom. It is bordered by the Tafila from the south and by Madaba governorate from the north. Karak governorate is 120 km away from the capital Amman. The idea of the project came because of the urban growth and expansion and the fact that the governorate does not have an automatic brick factory that produces large quantities in accurate measurements that meet the demands of the governorate and the Southern region.

The project, if implemented, will provide the bricks in sufficient quantities and high quality.

The following table shows the main expected results of the project:

Table1 : Economic Indicators

Total Investment cost (JD)	767,156
Internal Return Rate	20.52%
Payback Period (Years)	6
Net Present Value (JD)	332,719
Job Oppurtunities	18
Projected Annual Revenue for 10 years (JD)	7,190,719
Total Operating Cost for 10 years (JD)	5,252,568
Gross Profit for 10 years (JD)	627,911

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

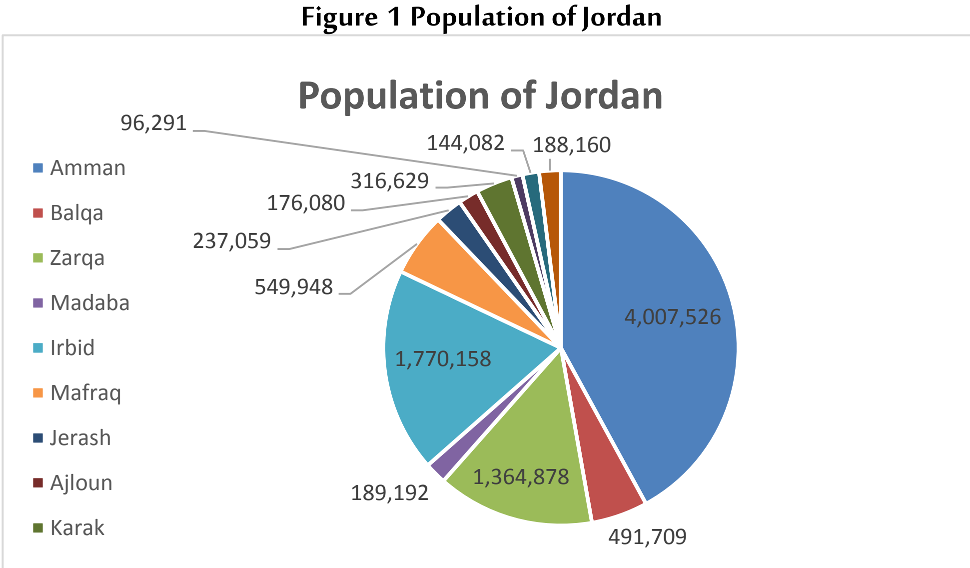
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

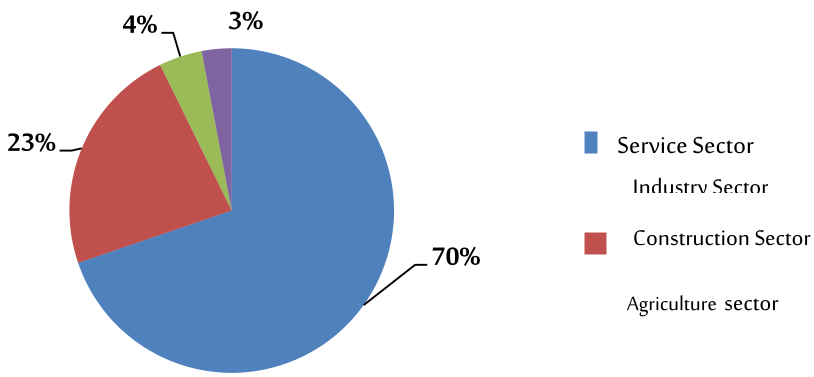
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.
Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.
The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The idea of the project is to establish an automatic brick factory that produces bricks automatically using machines to mix cement, sand and other necessary industrial materials with water in calculated proportions. The mixture is then pressed and molded to produce the bricks used in construction.

The idea of the project came because of the urban growth and expansion to the fact that the governorate does not contain an automatic brick factory that produces brick in large quantities and accurate measures.

Project Objectives

The Objectives of the project are summarized as follows:

- Achieve earnings and financial returns for the investors
- Provide bricks for construction works in Al-Karak
- Provide job opportunities

The Proposed Product

The proposed project provides bricks used for building purposes in quantities greater than the local production in the governorate and with high quality:

- Brick (20×10×40) cm
- Brick (20×15×40) cm
- Brick (20×20×40) cm

Target Segments

The target segments for the project will be the contractors and the residents of Karak governorate and those responsible for the construction and residential projects in the region.

Market Size Analysis

The number of buildings in Karak Governorate according to the Department of Statistics was 19,964, and 42,460 buildings in the southern governorates.

The following table shows the most common sectors using bricks:

Table 3: Top (5) Economic Sectors in using Bricks

No.	Economic Sectors	Percentage (%)
1	Constructions	98.36
2	Other building materials manufacturing	0.37
3	Bricks manufacturing	0.17
4	Other services	0.16
5	Works services	0.01
	Total sector production for intermediate consumption	**99.09**
	Total sector production of final demand components	**0.91**
	Total Production	**100**

Department of Statistics / Sub-Sector (Construction)

The following table shows construction activities in Jordan based on building licenses and construction area (1,000 sqm) during the period (2012-2016):

Table4 : Construction Activity in Jordan

Period	2012	2013	2014	2015	2016
Housing					
Licenses No.	**29,940**	**33,906**	**36,642**	**32,882**	**36,367**
Amman	8,429	10,063	11,118	10,496	11,293
Irbid	7,956	8,096	8,904	7,596	7,172
Zarqa	2,804	3,223	3,103	2,765	3,292
Other	10,751	12,524	13,517	12,025	14,610
Area (sqm)	**10,691.2**	**11,817.6**	**12,578.4**	**10,879.0**	**10,799.7**
Amman	6,013.2	6,075.6	6,084.1	4,967.3	4,019.7

Period	2012	2013	2014	2015	2016
Irbid	1,806.0	2,243.1	2,611.9	2,378.1	2,599.3
Zarqa	756.3	942.9	983.4	888.8	1,076.3
Other	2,115.7	2,556.0	2,899.0	2,644.8	3,104.3
Other Purposes					
Licenses No.	**2,526**	**2,695**	**2,936**	**2,893**	**3,043**
Amman	1,188	1,176	1,334	1,322	1,373
Irbid	431	568	672	657	547
Zarqa	338	423	471	477	600
Other	569	528	459	437	523
Area (sqm)	**2,215.5**	**2,167.5**	**2,413.9**	**2,243.8**	**2,510.5**
Amman	1,604.0	1,338.3	1,616.8	1,457.0	1,618.5
Irbid	131.1	254.8	233.6	268.9	188.6
Zarqa	146.6	175.6	265.2	217.3	294.4
Other	333.8	398.8	298.3	300.6	409.0

Source: Monthly Statistical Report / Central Bank 2016

- The area licensed for construction in 2016 in the governorates (Salt, Mafraq, Karak, Madaba, Jerash, Ajloun, Aqaba, Ma'an and Tafila) was 3,513,300 sqm.
- To calculate the area of external walls, (the total area multiplied by the number of walls (4), as each building has 4 outer walls.
- The area of all building walls is equal to 1,561,466.8 sqm.
- Windows and doors were assumed to make up 25% of total construction.
- Total area of walls excluding doors and windows is equal to 1,171,100 sqm.
- Since brick is used in the construction of internal walls in buildings, it was assumed that the proportion of interior walls is 25% of the total area of construction.
- The total building area used bricks, including interior walls (excluding windows and doors) is equal to 1,463,875 sqm.
- 12.5 Bricks are needed per square meter of construction.
- Thus, the number of bricks used in the total construction area is 18,298,438 bricks.

Competitors Analysis

There is no automatic brick factory in the Karak governorate, but there are a limited number of manual brick production workshops as shown in the following table:

Table5 : Manual Brick Factories in Al-Karak

No.	Owner's Name	Address
1	Issam Shaher Odeh Al-aghawat	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
2	Heirs of Mahmoud Mohammad Salem Al-karaki	Al-Herafyyeh
3	Ali Ahmad Mansour Al-Bawalez	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
4	Mohammad Haidar Odeh Abu-aisheh	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
5	Mohammad Sliman Saleem Al-Boshi	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
6	Mohammad Omar Taleb Al-mubaideen	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
7	Elias Naji Ameen Al-ja'afreh	Rakin Area / Main Street
8	Hayyat Mahmoud Ibrahim Al-s'oub	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
9	Mohammad Abdel-majeed Al-Dmour	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
10	Adel Salem Awwad Al-boshi	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
11	AbdelRahman Mohammad Ameen Al-bastanji	Manshieh area
12	Tawfiq Atallah Karim Aljafra	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
13	Samer Tawfiq Mitri Alkwalait	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7

No.	Owner's Name	Address
14	Waleed Fahim Nof Albayyadh	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
15	Bassem nedam Abdullah Rba'a	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
16	Othman Abdel Hamid Melhan Al - Maaitah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
17	Nasser Ghaleb Mahmoud Al-Karaki	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
18	Mohammed Lafi Mahmoud Abu Nawas	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
19	Kamal Hamad Ali Al - Maaitah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
20	Iyad Naim Salim Alqalanza Al-Hijazin	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
21	Salman Sulaiman Salman Al Oudat	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
22	Abdullah Karim Mahmoud and Mohammed Karim Al-Bayaydah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
23	Suliman Abdul Karim Muslim Jafra	Rakin Area
24	Muslih Yahya mousa Al-Karaki	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
25	Abdul Karim Ahmed Mohammed Abu Amr	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
26	The heirs of Izzat Abdul Aziz and his partner Hamada Mohammed Shafshi	Rakin Area
27	Abdul Majeed Ali Saleem Al - Maaitah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7

No.	Owner's Name	Address
28	Ibrahim Abdul Hamid Farhan Al - Jarajra	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
29	Fahd Yahya Mousa Al - Karaki	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
30	Abdulhameed Farhan Al - Jarajreh	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
31	Yassen Ibrahim Mohmel Al-Sohyimat	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
32	Ali Mahmoud Karim Al-Maaitah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
33	Raed Abdul Hamid Farhan Al - Jarajra	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
34	Omar Deeb Hassan Maali	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
35	Mohammed Karim Mahmood Albiyaideh and Bassam Mahmoud Al-	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
36	Bayaydeh & Karaki Company	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
37	Ali Khalife Abdoun Al - Maaitah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
38	Ayman Saber Itris Abdul-Ghani and his partner company	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7
39	Abdulqader Mohammed Hamed Al-Maaitah	Al-Lajoun Al-herafyyeh Al-jadedah Plot no.156 Basin no.7

Production Capacity

Based on the proposed production line, the annual output of the project is 2,500,000 pieces of brick.

Thus, the market share of the project is approximately 13.66% of the total market size in Karak governorate.

Raw Materials

The following table shows the number of bricks per cubic meter:

Table 6 : Number of Brick per One Cubic Meter

Bricks	(40×20×20) cm	(40×20×15) cm	(40×20×10) cm
One Cubic Meter	60 Bricks	75 Bricks	120 Bricks

The following table shows the average quantities required to produce 100 bricks of different measurements:

Table7 : Required Quantities of Raw Materials

Raw Material	Quantity (KG)	Cost JD / 100 Bricks
Cement	75	**7.5**
Gravel and Sand	1250	**7.5**

Price Analysis and Pricing Policy

After studying the local market, the selling prices were set as follows:

- Brick (40×10×20) cm at 0.24 JD.
- Brick (40×10×20) cm at 0.35 JD.
- Brick (40×10×20) cm at 0.40 JD.

The Projected Production Rate

The following table shows the projected production rate of the proposed project:

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production rate	60%	65%	70%	75%	80%	85%	90%	90%	95%	100%

Projected Revenues Analysis

The expected revenues for 10 years are as follows:

Table 8 : Revenues of the project during the years 2018-2027

Projected Revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues of (40×10×20) Brick	72000	80340	86520	92700	98880	105060	111240	111240	117420	123600
Revenues of (40×15×20) Brick	236250	263616	283894	304172	324450	344728	365006	365006	385284	405563
Revenues of (40×20×20) Brick	210000	234325	252350	270375	288400	306425	324450	324450	342475	360500
Total Revenues	518250	578281	622764	667247	711730	756213	800696	800696	845179	889663

The production capacity of the plant is divided according to the following percentages:

Table9 : Product percentage of Production Capacity

Product Percentage of Production Capacity	
(40×10×20) Brick	20%
(40×15×20) Brick	45%
(40×20×20) Brick	35%

4. Technical Analysis

Required Human Resources

The following table shows the human resources required for the proposed project:

Table 10 : Required Labor

Job Title	No.
Indirect Employees	
General Manager	1
Accountant	1
Total	2
Direct Employees	
Production Engineer	1
Technicians	2
Laboratory Worker	1
Driver	4
Workers	8
Total	16
The Total Number of Employees	**18**

The following is an explanation of the functions of the labor forces:

- **General Manager**: Propose the general policy of the plant, prepare and plans and follow up on their implementation; carry out technical and administrative supervision of staff, the development of procedures and ensure occupational health and safety requirements are met.
- **Accountant:** track financial matters and financial auditing.
- **Production Engineer**: Ensure that all products are manufactured efficiently and with high quality, according to the planned instructions using appropriate technology.
- **Technician**: Work on machines and follow-up on work line under the supervision of the production engineer and by implementing all instructions required..

- **Laboratory worker**: Check the quality of products and compare it with applicable standards.
- **Driver:** Driving vehicles to carry out the assigned delivery tasks.
- **Worker:** Prepare and process materials in required quantities and transfer products.

Site Analysis

The project will be on a large land far from residential areas, and provided with all necessary services required to complete the production process.

Technical Requirements Analysis

Land and construction works

The project will be constructed on an area of 3,000 sqm. The following table shows the required land:

Table11 : Required Land

Land	Area (sqm)
The required land	**3,000**

The required construction works for the project is represented as follows:

Table 12 : Required Construction Works

Construction Works	Area (sqm)
Buildings	500
Designing	500
Paving	100
others	-

Machinery and Equipment

The following table shows the machinery and equipment needed for the project:

Table13 : Required Machinery and Equipment

Machinery and Equipment	Quantity
Calibration machine	1
Materials mixing machine	1
Brick Manufacturing Machine	1
Treatment System	1
Conveyer	1
Laboratory Tools	1
Other	1

Furniture

The project needs furniture for administrative offices and a workers' lounge.

Vehicles and Transportation

The project needs Vehicles as follows:

Table14 : The Required Vehicles

Vehicles	Quantity
Heavy Truck	2
Medium transportation vehicle	2

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 15: General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 16: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 18: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	45.00%
Utilities Expenses from Total Revenues	3.00%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	2.00%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	0.50%

Table 19: Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	**Value**
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Table 20: Risk Premuim

Risk Premuim	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premuim (3)	10.12%
Income Tax Rate	5.00%
Beta (4)	1
Growth rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 21: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	306,863
Loans Value	460,294
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	460,294
Loans Percentage	%60
Equity	
Cost of Equity	16.6%
Equity Value	306,863
Equety Percentage	%40
Gross	
Project Value	767,156
Proportion of the project	%100

Capital Expenditures

The estimated cost of the project is 767,156 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 22: Land Capital Expenditure

Land	Cost(Jordan Dinar/Meters Square)	Area (M2)	Cost
Required Land	7	3,000	21,000
Total			**21,000**

Table 23: Construction Work Capital Expenditure

Description	Cost (JD per Meter Squared)	Area	Gross Cost (JD)
Construction Works	200		100,000
Designing	7		3,500
Paving	12		1,200
Others			2,000
Gross Total			**106,700**

Table 24: Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Calibration machine	1	53,200	53,200
Materials mixing machine	1	80,750	80,750
Brick Manufacturing Machine	1	104,500	104,500
Treatment System	1	71,250	71,250
Convyer	1	14,250	14,250
Labotary Tools	1	6,650	6,650
Others	1	19,000	19,000
Total (JD)			**349,600**

Table 25: Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	7,000
Total			7,000

Table 26: Transporation and Vehicles Capital Expenditure

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Heavy Trucks	2	30,000	60,000
Medium Trucks	2	23,000	46,000
Gross Total			106,000

Table 27: Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Govermental Fees	800
Transportation	1,500
Marketing	750
Training	500
Total	**3,550**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 28: Initial Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	92,594
General and administrative expenses	3	%25	2,323
Marketing expenses	3	%25	648
Indirect Salaries	3	%25	8,000
Total			**103,565**

Table 29: Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Land	21,000	10%	23,100	%3.0
Construction works	106,700	10%	117,370	%15.3
Machinery &Equipment	349,600	10%	384,560	%50.1
Furniture	7,000	10%	7,700	%1.0
Vehicles	106,000	10%	116,600	%15.2
Pre-Operating Expenses	3,550	10%	3,905	%0.5
Working capital	103,565	10%	113,921	%14.8
Total (JD)	**697,415**		**767,156**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (767,156) JD, where 60% of the project will be financed through loans (460,294) JD, and the remaining (40%) through (Equity) with (306,863) JD. The following table summarizes the general structure of the required financing:

Table 30: Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	306,863	%40
Loans	460,294	%60
Total	**767,156**	**%100**
Use of Fund	**Value**	**Ratio**
Capital Expenditures	649,330	%84.64
Pre-Operating Expenses	3,905	%0.51
Working Capital	113,921	%14.85
Total	**767,156**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 31: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	233,213	260,226	280,244	300,261	320,279	340,296	360,313	360,313	380,331	400,348
Utilities from Revenues	15,548	17,348	18,683	20,017	21,352	22,686	24,021	24,021	25,355	26,690
Maintenance from Revenues	5,183	5,783	6,228	6,672	7,117	7,562	8,007	8,007	8,452	8,897
Disposables from Revenues	10,365	11,566	12,455	13,345	14,235	15,124	16,014	16,014	16,904	17,793
Direct Salaries	93,600	97,438	101,433	105,591	109,921	114,427	119,119	124,003	129,087	134,379
Others	12,470	39,236	41,904	44,589	47,290	50,010	52,747	53,236	56,013	58,811
Total	**370,377**	**431,597**	**460,946**	**490,476**	**520,193**	**550,106**	**580,221**	**585,594**	**616,141**	**646,918**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 32: General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	500	515	530	546	563	580	597	615	633	652
Hospitality	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Staionary and Printing	600	618	637	656	675	696	716	738	760	783
Transportaton	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Training	500	515	530	546	563	580	597	615	633	652
Insurance	3,247	3,344	3,444	3,548	3,654	3,764	3,877	3,993	4,113	4,236
Miscellaneous	845	870	896	923	951	979	1,009	1,039	1,070	1,102
Total	**9,291**	**9,570**	**9,857**	**10,153**	**10,457**	**10,771**	**11,094**	**11,427**	**11,770**	**12,123**

Marketing Expenses

Table 33: Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	2,591	2,891	3,114	3,336	3,559	3,781	4,003	4,003	4,226	4,448

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 34: Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Accountant	500	521	542	564	587	611	636	662	690	718
Direct Employees										
Production Engineer	850	885	921	959	998	1,039	1,082	1,126	1,172	1,220
Technician	400	416	433	451	470	489	509	530	552	574
Laboratory Worker	400	416	433	451	470	489	509	530	552	574
Driver	350	364	379	395	411	428	445	464	483	502
Worker	300	312	325	338	352	367	382	397	414	431

Table 35: Expected yearly salaries for the years (2018-2027)

المسمى الوظيفي	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Accountant	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Total Indirect Salaries	**32,000**	**33,312**	**34,678**	**36,100**	**37,580**	**39,120**	**40,724**	**42,394**	**44,132**	**45,942**
Direct Employees										
Production Engineer	13,600	14,158	14,738	15,342	15,971	16,626	17,308	18,017	18,756	19,525
Technician	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Laboratory Worker	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Driver	22,400	23,318	24,274	25,270	26,306	27,384	28,507	29,676	30,893	32,159
Worker	38,400	39,974	41,613	43,319	45,096	46,945	48,869	50,873	52,959	55,130
Total Direct Salaries	**93,600**	**97,438**	**101,433**	**105,591**	**109,921**	**114,427**	**119,119**	**124,003**	**129,087**	**134,379**
Total Salaries	**125,600**	**130,750**	**136,110**	**141,691**	**147,500**	**153,548**	**159,843**	**166,397**	**173,219**	**180,321**

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Depreciations

:The following tables show all equipment costs, expenses and depreciation rates

Table 36: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	23,100	%0.0	%0.0
Construction Works	117,370	%5.0	%0.0
Machinaries	384,560	%10.0	%2.0
Furniture	7,700	%10.0	%5.0
Vehicles	116,600	%10.0	%0.0
	649,330		

Table 37: Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	649,330	657,406	665,655	674,082	682,690	691,483	700,466	709,643	719,020	728,600
Total Depreciation Expenses	56,755	57,562	58,387	59,230	60,090	60,970	61,868	62,786	63,723	64,682
Total Accumulated Depreciation	56,755	114,317	172,704	231,933	292,024	352,994	414,862	477,647	541,371	606,052
Total Additions	0	8,076	8,249	8,426	8,608	8,793	8,983	9,177	9,377	9,580
Total Net Book Values	592,576	543,090	492,952	442,149	390,666	338,489	285,604	231,996	177,649	122,548

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 38: Loan Details

Loan Value	460,294
Annual Interst Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	909,895
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				460,294
2019	119,816	43,636	76,180	384,114
2020	119,816	36,414	83,402	300,713
2021	119,816	28,508	91,308	209,405
2022	119,816	19,852	99,964	109,441
2023	119,816	10,375	109,441	0

Income Statement

Table 39: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	518,250	578,281	622,764	667,247	711,730	756,213	800,696	800,696	845,179	889,663
Gross Operating Revenues	518,250	578,281	622,764	667,247	711,730	756,213	800,696	800,696	845,179	889,663
Operating Expenses	-370,377	-431,597	-460,946	-490,476	-520,193	-550,106	-580,221	-585,594	-616,141	-646,918
Gross Operating Profit	147,873	146,684	161,817	176,771	191,537	206,108	220,475	215,103	229,038	242,745
Gross Profit Percentage	*%29*	*%25*	*%26*	*%26*	*%27*	*%27*	*%28*	*%27*	*%27*	*%27*
Salaries and Benefits (Indirect Staff)	-32,000	-33,312	-34,678	-36,100	-37,580	-39,120	-40,724	-42,394	-44,132	-45,942
General and Administraive Expenses	-9,291	-9,570	-9,857	-10,153	-10,457	-10,771	-11,094	-11,427	-11,770	-12,123
Markeing Expenses	-2,591	-2,891	-3,114	-3,336	-3,559	-3,781	-4,003	-4,003	-4,226	-4,448
Pre Operating Expenses	-3,905									
Gross Indirect Expenses	-47,788	-45,773	-47,649	-49,589	-51,596	-53,673	-55,822	-57,825	-60,128	-62,513
Income before Interest, Depreciation and Tax	100,085	100,910	114,169	127,182	139,941	152,435	164,653	157,278	168,910	180,232
Fixed Assets Depreciations	-56,755	-57,562	-58,387	-59,230	-60,090	-60,970	-61,868	-62,786	-63,723	-64,682
Income before Tax and Interests	43,331	43,348	55,782	67,953	79,851	91,465	102,785	94,492	105,187	115,550
Bank Interests	-43,636	-36,414	-28,508	-19,852	-10,375	-0	-0	-0	-0	-0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	-305	6,934	27,274	48,101	69,476	91,465	102,785	94,492	105,187	115,550
Income Tax	0	-331	-1,364	-2,405	-3,474	-4,573	-5,139	-4,725	-5,259	-5,778
Net Profit	-305	6,603	25,910	45,696	66,002	86,892	97,646	89,768	99,927	109,773
Net Profit Percentage	*%0*	*%1*	*%4*	*%7*	*%9*	*%11*	*%12*	*%11*	*%12*	*%12*
Compulsory Reserves	0	-660	-2,591	-4,570	-6,600	-8,689	-9,765	-8,977	-9,993	-10,977
Retained Earnings	-305	5,638	28,957	70,083	129,485	207,688	295,569	376,360	466,294	565,090

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 40: Expected Cashflows

العملة دينار أردني	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	-305	6,603	25,910	45,696	66,002	86,892	97,646	89,768	99,927	109,773
Non-monetary adjustments										
Bank benefits	43,636	36,414	28,508	19,852	10,375	0	0	0	0	0
Consumption	56,755	57,562	58,387	59,230	60,090	60,970	61,868	62,786	63,723	64,682
Operating cash flows before changes in	100,085	100,579	112,805	124,777	136,467	147,862	159,514	152,553	163,651	174,454

العملة دينار أردني	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Shortage (increase) in inventory	-19,434	-2,251	-1,668	-1,668	-1,668	-1,668	-1,668	0	-1,668	-1,668
Decrease (increase) in accounts receivable	0	0	0	0	0	0	0	0	0	0
(Decrease) increase in accounts payable	23,318	5,815	1,586	1,617	1,649	1,682	1,717	913	1,790	1,829
Change in working capital	3,884	3,564	-82	-51	-19	14	49	913	122	161
Net cash flows (used) from operating	103,969	104,143	112,723	124,726	136,448	147,876	159,562	153,467	163,773	174,615
Cash flows from investing activities										
(Purchase) of fixed assets	-649,330	-8,076	-8,249	-8,426	-8,608	-8,793	-8,983	-9,177	-9,377	-9,580
Net flows (used) from investment	-649,330	-8,076	-8,249	-8,426	-8,608	-8,793	-8,983	-9,177	-9,377	-9,580
Cash flows from financing activities										
Equity capital	306,863									
Loan Amortization	-76,180	-83,402	-91,308	-99,964	-109,441	0	0	0	0	0
Bank Interest Rate	-43,636	-36,414	-28,508	-19,852	-10,375	-0	-0	-0	-0	-0
Loans	460,294	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing	647,341	-119,816	-119,816	-119,816	-119,816	0	0	0	0	0
Net increase (decrease) in cash	101,980	-23,749	-15,341	-3,516	8,025	139,083	150,579	144,289	154,396	165,035
Cashflows at the Beginning of Period	0	101,980	78,231	62,889	59,374	67,399	206,481	357,060	501,350	655,746
Cashflows at the End of Period	101,980	78,231	62,889	59,374	67,399	206,481	357,060	501,350	655,746	820,781

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 41: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	101,980	78,231	62,889	59,374	67,399	206,481	357,060	501,350	655,746	820,781
Inventory	19,434	21,686	23,354	25,022	26,690	28,358	30,026	30,026	31,694	33,362
accounts receivable	0	0	0	0	0	0	0	0	0	0
Total assets traded	**121,414**	**99,916**	**86,243**	**84,395**	**94,089**	**234,839**	**387,086**	**531,376**	**687,440**	**854,143**
Non-current assets										
Fixed assets (net)	592,576	543,090	492,952	442,149	390,666	338,489	285,604	231,996	177,649	122,548
Total non-current assets	**592,576**	**543,090**	**492,952**	**442,149**	**390,666**	**338,489**	**285,604**	**231,996**	**177,649**	**122,548**
Total assets	**713,990**	**643,006**	**579,195**	**526,544**	**484,754**	**573,328**	**672,690**	**763,371**	**865,089**	**976,691**
Current liabilities										
Payables	23,318	29,133	30,719	32,337	33,985	35,668	37,384	38,298	40,088	41,917
Remaining amount of Loan	83,402	91,308	99,964	109,441	0-	0-	0-	0-	0-	0
Total current liabilities	**106,720**	**120,441**	**130,683**	**141,777**	**33,985**	**35,668**	**37,384**	**38,298**	**40,088**	**41,917**

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	300,713	209,405	109,441	0	0	0	0	0	0	0
Total long - term liabilities	**300,713**	**209,405**	**109,441**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total liabilities	**407,432**	**329,846**	**240,124**	**141,777**	**33,985**	**35,668**	**37,384**	**38,298**	**40,088**	**41,917**
Property rights										
Members Contributions	306,863	306,863	306,863	306,863	306,863	306,863	306,863	306,863	306,863	306,863
Compulsory reserve	0	660	3,251	7,821	14,421	23,110	32,875	41,852	51,844	62,822
Profit (loss)	305-	5,638	28,957	70,083	129,485	207,688	295,569	376,360	466,294	565,090
Total equity	306,558	313,160	339,071	384,767	450,769	537,661	635,306	725,074	825,001	934,774
Total liabilities and equity	**713,990**	**643,006**	**579,195**	**526,544**	**484,754**	**573,328**	**672,690**	**763,371**	**865,089**	**976,691**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 42: Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	القيمة
Net Free Cashflows	-767,156	107,874	96,067	104,47	116,30	127,84	139,08	150,57	144,28	154,39	165,03	2,131,703
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash Flows	**-767,156**	**76,677**	**60,940**	**59,145**	**58,758**	**57,642**	**55,966**	**54,076**	**46,244**	**44,161**	**42,127**	**544,139**

Table below (43) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 43: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-767,156	107,874	96,067	104,474	116,300	127,841	139,083	150,579	144,289	154,396	165,035
Project Value Returned	767,156	659,282	563,216	458,742	342,442	214,601	75,519	-75,060	-219,350	-373,746	-538,781
Payback Period (Year)	6	1	1	1	1	1	1	0	0	0	0

Table 44: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	332,719
Payback Period	6
Internal Rate of Return	%20.52

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 45: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%20.52	6	%12.05
Revenues declined by 10%	%16.35	8	12.05%
Operating Expenses Increased by 10%	%12.83	9	12.05%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 332,719 Jordan Dinars considering that the project provides 18 Job Opportunitites for the governorate residents.